

Last login: Thu Jul 2 11:18:42 on ttys003
adityadave@Adityas-MacBook-Air-3 ~ % psql -d tpch-200mb
psql (18.4)
Type "help" for help.

tpch-200mb=# SET ROLE alice;

EXPLAIN (ANALYZE, VERBOSE, BUFFERS)

```
select
  s_name,
  s_address
from
  supplier,
  nation
where
  s_suppkey in (
    select
      ps_suppkey
    from
      partsupp_secure
    where
      ps_partkey in (
        select
          p_partkey
        from
          part
        where
          p_name like '%ivory%'
      )
    and ps_availqty > (
      select
        0.5 * sum(l_quantity)
      from
        lineitem
      where
        l_partkey = ps_partkey
        and l_suppkey = ps_suppkey
        and l_shipdate >= date '1995-01-01'
        and l_shipdate < date '1996-01-01' + interval '1' year
      )
    )
  and s_nationkey = n_nationkey
  and n_name = 'FRANCE'
order by
  s_name;
```

-- Planning Time: 12.403 ms

-- Execution Time: 814.154 ms

SET

WARNING: Log 1: Planner Input:

DETAIL: {QUERY

```
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks true
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:ctelist <>
:rttable (
  {RANGETBLENTRY
  :alias <>
  :eref
  {ALIAS
  :aliasname supplier
  :colnames ("s_suppkey" "s_name" "s_address" "s_nationkey" "s_phone" "
    s_acctbal" "s_comment")
  }
  :rtekind 0
  :relid 17399
  :inh true
  :relkind r
  :rellockmode 1
  :perminfoindex 1
  :tablesample <>
  :lateral false
  :inFromCl true
  :securityQuals <>
  }
  {RANGETBLENTRY
  :alias <>
  :eref
  {ALIAS
  :aliasname nation
  :colnames ("n_nationkey" "n_name" "n_regionkey" "n_comment")
  }
  :rtekind 0
  :relid 17376
  :inh true
  :relkind r
  :rellockmode 1
  :perminfoindex 2
  :tablesample <>
  :lateral false
  :inFromCl true
  :securityQuals <>
  }
  )
:rtperminfos (
  {RTEPERMISSIONINFO
  :relid 17399
  :inh true
  :requiredPerms 2
  :checkAsUser 0
  :selectedCols (b 8 9 10 11)
  :insertedCols (b)
  :updatedCols (b)
  }
  {RTEPERMISSIONINFO
  :relid 17376
  :inh true
  :requiredPerms 2
  :checkAsUser 0
  :selectedCols (b 8 9)
  :insertedCols (b)
  :updatedCols (b)
  }
  )
)
```

```
:rtperminfos (
  {RTEPERMISSIONINFO
  :relid 17399
  :inh true
  :requiredPerms 2
  :checkAsUser 0
  :selectedCols (b 8 9 10 11)
  :insertedCols (b)
  :updatedCols (b)
  }
  {RTEPERMISSIONINFO
  :relid 17376
  :inh true
  :requiredPerms 2
  :checkAsUser 0
  :selectedCols (b 8 9)
  :insertedCols (b)
  :updatedCols (b)
  }
  )
)
```

```
:jointree
{FROMEXPR
:fromlist (
  {RANGETBLREF
  :rtindex 1
  }
  {RANGETBLREF
  :rtindex 2
  }
  )
:quals
{BOOLEXP
:boolop and
:args (
  {SUBLINK
  :subLinkType 2
  :subLinkId 0
  :testexpr
  {OPEXP
  :opno 96
```

```

:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
  {VAR
    :varno 1
    :varattno 1
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 1
    :location 129
  }
  {PARAM
    :paramkind 2
    :paramid 1
    :paramtype 23
    :paramtypmod -1
    :paramcollid 0
    :location -1
  }
)
:location 139
}
:operName ("=")
:subselect
{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks true
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:ctelist <>
:rtable (
  {RANGETBLENTRY
    :alias <>
    :eref
      {ALIAS
        :aliasname partsupp_secure
        :colnames ("ps_partkey" "ps_suppkey" "ps_availqty" "ps_su
          pplycost" "ps_comment")
      }
    :rtekind 1
  }
  {SUBQUERY
    {QUERY
      :commandType 1
      :querySource 0
      :canSetTag true
      :utilityStmt <>
      :resultRelation 0
      :hasAggs false
      :hasWindowFuncs false
      :hasTargetSRFs false
      :hasSubLinks true
      :hasDistinctOn false
      :hasRecursive false
      :hasModifyingCTE false
      :hasForUpdate false
      :hasRowSecurity false
      :hasGroupRTE false
      :isReturn false
      :ctelist <>
    }
    :rtable (
      {RANGETBLENTRY
        :alias
          {ALIAS
            :aliasname ps
            :colnames <>
          }
        :eref
          {ALIAS
            :aliasname ps
            :colnames ("ps_partkey" "ps_suppkey" "ps_availqty"
              "ps_supplycost" "ps_comment")
          }
        :rtekind 0
        :relid 17409
        :inh true
        :relkind F
        :rellockmode 1
        :perminfoindex 1
        :tablesample <>
        :lateral false
        :inFromCl true
        :securityQuals <>
      }
    )
    :rteperminfos (
      {RTEPERMISSIONINFO
        :relid 17409
        :inh true
        :requiredPerms 2
        :checkAsUser 10
        :selectedCols (b 8 9 10 11 12)
        :insertedCols (b)
        :updatedCols (b)
      }
    )
  }
  {JOINTREE
    {FROMEXPR
      :fromlist (
        {RANGETBLREF
          :rtindex 1
        }
      )
    }
    :quals
      {BOOLEXP
        :boolop and
        :args (
          {OPEXP
            :opno 521
            :opfuncid 147
            :opresulttype 16
            :opretset false
            :opcollid 0
            :inputcollid 0
            :args (
              {VAR
                :varno 1
                :varattno 3
                :vartype 23
                :vartypmod -1
                :varcollid 0
                :varnullingrels (b)
                :varlevelsup 0
                :varreturningtype 0
                :varnosyn 1
                :varattnosyn 3
                :location -1
              }
            )
          }
        )
      }
    }
  }
  {CONST

```

```

:consttype 23
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location -1
:constvalue 4 [ 100 0 0 0 0 0 0 ]
}
)
:location -1
}
}
{SUBLINK
:sublinkType 0
:subLinkId 0
:testexpr <>
:operName <>
:subselect
{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:ctelist <>
:rtable (
{RANGETBENTRY
:alias
{ALIAS
:aliasname s
:colnames <>
}
:eref
{ALIAS
:aliasname s
:colnames ("s_suppkey" "s_name" "s_addr"
"ess" "s_nationkey" "s_phone" "s_acctbal"
"s_comment")
}
:rtekind 0
:relid 17399
:inh true
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
)
:rteperminfos (
{RTEPERMISSIONINFO
:relid 17399
:inh true
:requiredPerms 2
:checkAsUser 10
:selectedCols (b 0 13)
:insertedCols (b)
:updatedCols (b)
}
)
)
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
)
)
:quals
{BOOLEXPR
:boolop and
:args (
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location -1
}
{VAR
:varno 1
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 1
:varreturningtype 0
:varnosyn 1
:varattnosyn 2
:location -1
}
)
}
:location -1
}
{OPEXPR
:opno 1756
:opfuncid 1720
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 6
:vartype 1700
:vartypmod 983046
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location -1
}
}
{FUNCEXPR
:funcid 1740
:funcresulttype 1700
:funcset false
:funcvariadic false
:funcformat 2
}
}
}

```

```

                :funccollid 0
                :inputcollid 0
                :args (
                    {CONST
                        :consttype 23
                        :consttypmod -1
                        :constcollid 0
                        :constlen 4
                        :constbyval true
                        :constisnull false
                        :location -1
                        :constvalue 4 [ 0 0 0 0 0 0 0 ]
                    }
                )
                :location -1
            }
        )
        :location -1
    }
}

:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
    {TARGETENTRY
        :expr
        {CONST
            :consttype 23
            :consttypmod -1
            :constcollid 0
            :constlen 4
            :constbyval true
            :constisnull false
            :location -1
            :constvalue 4 [ 1 0 0 0 0 0 0 0 ]
        }
        :resno 1
        :resname ?column?
        :ressortgroupref 0
        :resorigtbl 0
        :resorigcol 0
        :resjunk false
    }
)
:override 0
:onConflict <>
:ReturningOldAlias <>
:ReturningNewAlias <>
:ReturningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}
:location -1
}
}
:location -1
}
}

:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
    {TARGETENTRY
        :expr
        {VAR
            :varno 1
            :varattno 1
            :vartype 23
            :vartypmod -1
            :varcollid 0
            :varnullingrels (b)
            :varlevelsup 0
            :varreturningtype 0
            :varnosyn 1
            :varattnosyn 1
            :location -1
        }
        :resno 1
        :resname ps_partkey
        :ressortgroupref 0
        :resorigtbl 17409
        :resorigcol 1
        :resjunk false
    }
)
{TARGETENTRY
:expr
{VAR
    :varno 1
    :varattno 2
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 2
    :location -1
}
:resno 2
:resname ps_suppkey
:ressortgroupref 0
:resorigtbl 17409
:resorigcol 2
:resjunk false
}
{TARGETENTRY
:expr
{VAR
    :varno 1
    :varattno 3
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 3
    :location -1
}
:resno 3
:resname ps_availqty
:ressortgroupref 0
:resorigtbl 17409
:resorigcol 3
:resjunk false
}
{TARGETENTRY
:expr

```

```

        {VAR
        :varno 1
        :varattno 4
        :vartype 1700
        :vartypmod 983046
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 1
        :varattnosyn 4
        :location -1
        }
        :resno 4
        :resname ps_supplycost
        :ressortgroupref 0
        :resorigtbl 17409
        :resorigcol 4
        :resjunk false
    }
    {TARGETENTRY
    :expr
    {VAR
    :varno 1
    :varattno 5
    :vartype 1043
    :vartypmod 203
    :varcollid 100
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 5
    :location -1
    }
    :resno 5
    :resname ps_comment
    :ressortgroupref 0
    :resorigtbl 17409
    :resorigcol 5
    :resjunk false
    }
    )
    :override 0
    :onConflict <>
    :returningOldAlias <>
    :returningNewAlias <>
    :returningList <>
    :groupClause <>
    :groupDistinct false
    :groupingSets <>
    :havingQual <>
    :windowClause <>
    :distinctClause <>
    :sortClause <>
    :limitOffset <>
    :limitCount <>
    :limitOption 0
    :rowMarks <>
    :setOperations <>
    :constraintDeps <>
    :withCheckOptions <>
    :stmt_location -1
    :stmt_len -1
    }
    :security_barrier true
    :relid 19883
    :inh false
    :relkind v
    :rellockmode 1
    :perminfoindex 1
    :lateral false
    :inFromCl true
    :securityQuals <>
    }
    )
    :rteperminfos (
    {RTEPERMISSIONINFO
    :relid 19883
    :inh true
    :requiredPerms 2
    :checkAsUser 0
    :selectedCols (b 8 9 10)
    :insertedCols (b)
    :updatedCols (b)
    }
    )
    )
    :jointree
    {FROMEXPR
    :fromlist (
    {RANGETBLREF
    :rtindex 1
    }
    )
    )
    :quals
    {BOOLEXP
    :boolop and
    :args (
    {SUBLINK
    :subLinkType 2
    :subLinkId 0
    :testexpr
    {OPEXP
    :opno 96
    :opfuncid 65
    :opresulttype 16
    :opretset false
    :opcollid 0
    :inputcollid 0
    :args (
    {VAR
    :varno 1
    :varattno 1
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 1
    :location 309
    }
    {PARAM
    :paramkind 2
    :paramid 1
    :paramtype 23
    :paramtypmod -1
    :paramcollid 0
    :location -1
    }
    )
    }
    :location 320
    }
    )
    :operName ("=")
    :subselect
    {QUERY
    :commandType 1
    :querySource 0
    :canSetTag true
    :utilityStmt <>
    :resultRelation 0
    :hasAggs false
    :hasWindowFuncs false
    :hasTargetSRFs false
    :hasSublinks false
    :hasDistinctOn false
    :hasRecursive false

```

```

:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable (
  {RANGETBLENTY
    :alias <>
    :eref
      {ALIAS
        :aliasname part
        :colnames ("p_partkey" "p_name" "p_mfg" "p_b
          rand" "p_type" "p_size" "p_container" "p_reta
            ilprice" "p_comment")
        }
    :rtekind 0
    :relid 17387
    :inh true
    :relkind r
    :rellockmode 1
    :perminfoindex 1
    :tablesample <>
    :lateral false
    :inFromCl true
    :securityQuals <>
  }
)
:rteperminfos (
  {RTEPERMISSIONINFO
    :relid 17387
    :inh true
    :requiredPerms 2
    :checkAsUser 0
    :selectedCols (b 8 9)
    :insertedCols (b)
    :updatedCols (b)
  }
)
:jointree
  {FROMEXPR
    :fromlist (
      {RANGETBLREF
        :rtindex 1
      }
    )
    :quals
      {OPEXPR
        :opno 1209
        :opfuncid 850
        :opresulttype 16
        :opretset false
        :opcollid 0
        :inputcollid 100
        :args (
          {RELABELTYPE
            :arg
              {VAR
                :varno 1
                :varattno 2
                :vartype 1043
                :vartypmod 59
                :varcollid 100
                :varnullingrels (b)
                :varlevelsup 0
                :varreturningtype 0
                :varnosyn 1
                :varattnosyn 2
                :location 574
              }
            :resulttype 25
            :resulttypmod -1
            :resultcollid 100
            :relabelformat 2
            :location -1
          }
          {CONST
            :consttype 25
            :consttypmod -1
            :constcollid 100
            :constlen -1
            :constbyval false
            :constisnull false
            :location 586
            :constvalue 11 [ 44 0 0 0 37 105 118 111 1
              14 121 37 ]
          }
        )
        :location 581
      }
    }
  :mergeActionList <>
  :mergeTargetRelation 0
  :mergeJoinCondition <>
  :targetList (
    {TARGETENTRY
      :expr
        {VAR
          :varno 1
          :varattno 1
          :vartype 23
          :vartypmod -1
          :varcollid 0
          :varnullingrels (b)
          :varlevelsup 0
          :varreturningtype 0
          :varnosyn 1
          :varattnosyn 1
          :location 404
        }
        :resno 1
        :resname p_partkey
        :ressortgroupref 0
        :resorigtbl 17387
        :resorigcol 1
        :resjunk false
      }
    )
  :override 0
  :onConflict <>
  :returningOldAlias <>
  :returningNewAlias <>
  :returninglist <>
  :groupClause <>
  :groupDistinct false
  :groupingSets <>
  :havingQual <>
  :windowClause <>
  :distinctClause <>
  :sortClause <>
  :limitOffset <>
  :limitCount <>
  :limitOption 0
  :rowMarks <>
  :setOperations <>
  :constraintDeps <>
  :withCheckOptions <>
  :stmt_location 0
  :stmt_len 0
}
:location 320
}
{OPEXPR
:opno 1756
:opfuncid 1720
:opresulttype 16
:opretset false
:opcollid 0

```

```

:inputcollid 0
:args (
{FUNCEXP
:funcid 1740
:funcresulttype 1700
:funcresultset false
:funcvariadic false
:funcformat 2
:funccollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 3
:varatype 23
:varitypmode -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 3
:location 650
}
)
:location -1
}
{SUBLINK
:sublinkType 4
:sublinkid 0
:testexpr <>
:operName <>
:subselect
{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs true
:hasWindowFuncs false
:hasTargetSRFs false
:hasSublinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:ctelist <>
:rttable (
{RANGETBENTRY
:alias <>
:eref
{ALIAS
:aliasname lineitem
:colnames ("l_orderkey" "l_partkey" "l_sup
pkey" "l_linenum" "l_quantity" "l_exten
dedprice" "l_discount" "l_tax" "l_returnfl
ag" "l_linestatus" "l_shipdate" "l_commitd
ate" "l_receiptdate" "l_shipinstruct" "l_s
hipmode" "l_comment")
}
}
:rtkind 0
:relid 17440
:inh true
:relkind r
:rellockmode 1
:permindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
)
:rteperminfos (
{RTEPERMISSIONINFO
:relid 17440
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 9 10 12 18)
:insertedCols (b)
:updatedCols (b)
}
)
:jointree
{FROMEXP
:fromlist (
{RANGETBLREF
:rtindex 1
}
)
:quals
{BOOLEXP
:boolop and
:args (
{OPEXP
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 2
:varatype 23
:varitypmode -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 2
:location 931
}
{VAR
:varno 1
:varattno 1
:varatype 23
:varitypmode -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 1
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location 943
}
}
:location 941
}
{OPEXP
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 3
:varatype 23
:varitypmode -1

```

```

:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 3
:location 998
}
{VAR
:varno 1
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 1
:varreturningtype 0
:varnosyn 1
:varattnosyn 2
:location 1010
}
}
:location 1008
}
{OPEXPR
:opno 1098
:opfuncid 1090
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 11
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 11
:location 1065
}
{CONST
:consttype 1082
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location 1084
:constvalue 4 [ -34 -8 -1 -1 -1 -1 -1 -1 ]
}
)
:location 1076
}
{OPEXPR
:opno 2345
:opfuncid 2338
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 11
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 11
:location 1141
}
{OPEXPR
:opno 1076
:opfuncid 2071
:opresulttype 1114
:opretset false
:opcollid 0
:inputcollid 0
:args (
{CONST
:consttype 1082
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location 1159
:constvalue 4 [ -34 -8 -1 -1 -1 -1 -1 -1 ]
}
{CONST
:consttype 1186
:consttypmod 327679
:constcollid 0
:constlen 16
:constbyval false
:constisnull false
:location 1183
:constvalue 16 [ 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 ]
}
)
:location 1172
}
}
:location 1152
}
}
:location 994
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{OPEXPR
:opno 1760
:opfuncid 1726
:opresulttype 1700
:opretset false
:opcollid 0
:inputcollid 0
:args (
{CONST
:consttype 1700
:consttypmod -1
:constcollid 0
:constlen -1
:constbyval false
:constisnull false
:location 745
:constvalue 8 [ 32 0 0 0 -1 -128 -120 1 9 ]
}
}
{AGGREF
:aggfnoid 2114
:aggtype 1700

```

```

:aggcollid 0
:inputcollid 0
:aggtranstype 0
:aggargtypes (0 1700)
:aggdirectargs <>
:args (
  {TARGETENTRY
    :expr
    {VAR
      :varno 1
      :varattno 5
      :vartype 1700
      :vartypmod 983046
      :varcollid 0
      :varnullingrels (b)
      :varlevelsup 0
      :varreturningtype 0
      :varnosyn 1
      :varattnosyn 5
      :location 755
    }
    :resno 1
    :resname <>
    :ressortgrouppref 0
    :resorigtbl 0
    :resorigcol 0
    :resjunk false
  }
)
:aggorder <>
:aggdistinct <>
:aggfilter <>
:aggstar false
:aggvariadic false
:aggkind n
:aggpresorted false
:agglevelsup 0
:aggsplit 0
:aggno -1
:aggtransno -1
:location 751
}
)
:location 749
}
:resno 1
:resname ?column?
:ressortgrouppref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 0
}
:location 664
}
)
:location 662
}
)
:location 646
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
  {TARGETENTRY
    :expr
    {VAR
      :varno 1
      :varattno 2
      :vartype 23
      :vartypmod -1
      :varcollid 0
      :varnullingrels (b)
      :varlevelsup 0
      :varreturningtype 0
      :varnosyn 1
      :varattnosyn 2
      :location 191
    }
    :resno 1
    :resname ps_supkey
    :ressortgrouppref 0
    :resorigtbl 1983
    :resorigcol 2
    :resjunk false
  }
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 0
}
:location 139
}
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
  {VAR
    :varno 1
    :varattno 4
    :vartype 23
    :vartypmod -1

```

```

:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 4
:location 1240
}
{VAR
:varno 2
:varattno 1
:varatype 23
:varattpmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 1
:location 1254
}
}
:location 1252
}
{OPEXPR
:opno 1054
:opfuncid 1048
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 100
:args (
{VAR
:varno 2
:varattno 2
:varatype 1042
:varattpmod 29
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 2
:location 1278
}
{CONST
:consttype 1042
:consttppmod -1
:constcollid 100
:constlen -1
:constbyval false
:constisnull false
:location 1287
:constvalue 10 [ 40 0 0 0 70 82 65 78 67 69 ]
}
)
:location 1285
}
)
:location 1236
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 2
:varatype 1042
:varattpmod 29
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 2
:location 51
}
:resno 1
:resname s_name
:ressortgroupref 1
:resorigtbl 17399
:resorigcol 2
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 3
:varatype 1043
:varattpmod 44
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 3
:location 67
}
:resno 2
:resname s_address
:ressortgroupref 0
:resorigtbl 17399
:resorigcol 3
:resjunk false
}
}
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returninglist <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause (
{SORTGROUPCLAUSE
:tleSortGroupRef 1
:eqop 1054
:sortop 1058
:reverse_sort false
:nulls_first false
:hashable true
}
)
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 0
}

```

```

WARNING: max_parallel_hazard_walker: Param paramid=1 in_safe_list=0 → reason=outer_reference
WARNING: max_parallel_hazard_walker: Param paramid=1 in_safe_list=0 → reason=outer_reference
WARNING: max_parallel_hazard_walker: FuncExpr funcid=1740 proparallel=s (s=safe r=restricted u=unsafe)
WARNING: max_parallel_hazard_walker: FuncExpr funcid=1740 proparallel=s (s=safe r=restricted u=unsafe)
WARNING: Log 2 : After pull up At starting of Path Generation:

```

```

DETAIL: {QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs true
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable (
  {RANGETBLENTY
    :alias <>
    :eref
      {ALIAS
        :aliasname lineitem
        :colnames ("l_orderkey" "l_partkey" "l_suppkey" "l_linenumber" "l_quantity" "l_extendedprice" "l_discount" "l_tax" "l_returnflag" "l_linesatus" "l_shipdate" "l_commitdate" "l_receiptdate" "l_shipinstruct" "l_shipmode" "l_comment")
      }
    :rtckind 0
    :relid 17440
    :inh false
    :relkind r
    :rellockmode 1
    :perminfoindex 1
    :tablesample <>
    :lateral false
    :inFromCl true
    :securityQuals <>
  }
)
:rteperminfos (
  {RTEPERMISSIONINFO
    :relid 17440
    :inh true
    :requiredPerms 2
    :checkAsUser 0
    :selectedCols (b 9 10 12 18)
    :insertedCols (b)
    :updatedCols (b)
  }
)
:jointree
{FROMEXPR
  :fromlist (
    {RANGETBLREF
      :rtindex 1
    }
  )
}
:quals (
  {OEXPR
    :opno 96
    :opfuncid 65
    :opresulttype 16
    :opretset false
    :opcollid 0
    :inputcollid 0
    :args (
      {VAR
        :varno 1
        :varattno 2
        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 1
        :varattnosyn 2
        :location 931
      }
      {PARAM
        :paramkind 1
        :paramid 0
        :paramtype 23
        :paramtypmod -1
        :paramcollid 0
        :location 943
      }
    )
    :location 941
  }
  {OEXPR
    :opno 96
    :opfuncid 65
    :opresulttype 16
    :opretset false
    :opcollid 0
    :inputcollid 0
    :args (
      {VAR
        :varno 1
        :varattno 3
        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 1
        :varattnosyn 3
        :location 998
      }
      {PARAM
        :paramkind 1
        :paramid 1
        :paramtype 23
        :paramtypmod -1
        :paramcollid 0
        :location 1010
      }
    )
    :location 1008
  }
  {OEXPR
    :opno 1098
    :opfuncid 1090
    :opresulttype 16
    :opretset false
    :opcollid 0
    :inputcollid 0
    :args (
      {VAR
        :varno 1
        :varattno 11
        :vartype 1082
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 1
        :varattnosyn 11
        :location 1065
      }
      {CONST
        :consttype 1082
        :consttypmod -1

```

```

:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location 1084
:constvalue 4 [ -34 -8 -1 -1 -1 -1 -1 ]
}
)
:location 1076
}
{OPEXPR
:opno 2345
:opfuncid 2338
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 11
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 11
:location 1141
}
{CONST
:consttype 1114
:consttypmod -1
:constcollid 0
:constlen 8
:constbyval true
:constisnull false
:location -1
:constvalue 8 [ 0 32 -39 -79 49 -115 -1 -1 ]
}
)
:location 1152
}
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{OPEXPR
:opno 1760
:opfuncid 1726
:opresulttype 1700
:opretset false
:opcollid 0
:inputcollid 0
:args (
{CONST
:consttype 1700
:consttypmod -1
:constcollid 0
:constlen -1
:constbyval false
:constisnull false
:location 745
:constvalue 8 [ 32 0 0 0 -1 -128 -120 19 ]
}
{AGGREF
:aggfnoid 2114
:aggtype 1700
:aggcollid 0
:inputcollid 0
:aggtransype 0
:aggargtypes (o 1700)
:aggdirectargs <>
:args (
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 5
:vartype 1700
:vartypmod 983046
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 5
:location 755
}
:resno 1
:resname <>
:resortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:aggorder <>
:aggdistinct <>
:aggfilter <>
:aggstar false
:aggvariadic false
:aggkind n
:aggpresorted false
:agglevelsup 0
:aggsplit 0
:aggno -1
:aggtransno -1
:location 751
}
)
:location 749
}
}
:resno 1
:resname ?column?
:resortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
}
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 0
}

```

```
WARNING: is_parallel_safe: INPUT qual = (({RESTRICTINFO :clause {OPEXPR :opno 1098 :opfuncid 1090 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args {(VAR :varno 1 :varattno 11 :vartype 1
082 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 11 :location -1} {CONST :consttype 1082 :consttypmod -1 :constcollid 0 :constlen 4 :constby
val true :constisnull false :location -1 :constvalue 4 [ -34 -8 -1 -1 -1 -1 -1 ]}) :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof
false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_relds (b 1) :required_relds (b 1) :incompatible_relds (b) :outer_relds (b) :left_relds (b 1) :right_relds (b) :orclause <> :rinfo_
serial 3 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_buck
etsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hashseqoperator 0 :right_hashseqoperator 0} {RESTRICTINFO :clause {OPEXPR :opno 2345 :opfuncid 2338 :opresulttype 16 :opretset false :opcollid 0 :inputco
llid 0 :args {(VAR :varno 1 :varattno 11 :vartype 1082 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 11 :location -1} {CONST :consttype 1114
:consttypmod -1 :constcollid 0 :constlen 8 :constbyval true :constisnull false :location -1 :constvalue 8 [ 0 32 -39 -79 49 -115 -1 -1 ]}) :location -1} :is_pushed_down true :can_join false :pseudoconst
ant false :has_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_relds (b 1) :required_relds (b 1) :incompatible_relds (b) :outer_relds (b) :left_
_relds (b 1) :right_relds (b) :orclause <> :rinfo_serial 4 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies <> :left_em <> :right_em <> :outer_is_left fals
e :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hashseqoperator 0 :right_hashseqoperator 0} {RESTRICTINFO :clause {OPEXPR :opno 96 :opfuncid 65 :opr
esulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args {(VAR :varno 1 :varattno 2 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varatt
nosyn 2 :location -1} {PARAM :paramkind 1 :paramid 0 :paramtype 23 :paramtypmod -1 :paramcollid 0 :location -1}) :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false
:is_clone false :leakproof false :has_volatile 2 :security_level 0 :num_base_rels 1 :clause_relds (b 1) :required_relds (b 1) :incompatible_relds (b) :outer_relds (b 1) :left_relds (b 1) :right_reld
s (b) :orclause <> :rinfo_serial 1 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies (o 1976) :left_em {EQUIVALENCEMEMBER :em_expr {(VAR :varno 1 :varattno 2 :
vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1} :constvalue 4 [ -34 -8 -1 -1 -1 -1 -1 ]}) :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproo
f false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_relds (b 1) :required_relds (b 1) :incompatible_relds (b) :outer_relds (b) :left_relds (b 1) :right_relds (b) :orclause <> :rinfo_
serial 3 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_buc
ketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hashseqoperator 0 :right_hashseqoperator 0} {RESTRICTINFO :clause {OPEXPR :opno 2345 :opfuncid 2338 :opresulttype 16 :opretset false :opcollid 0 :inputc
ollid 0 :args {(VAR :varno 1 :varattno 11 :vartype 1082 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 11 :location -1} {CONST :consttype 1114
:consttypmod -1 :constcollid 0 :constlen 8 :constbyval true :constisnull false :location -1 :constvalue 8 [ 0 32 -39 -79 49 -115 -1 -1 ]}) :location -1} :is_pushed_down true :can_join false :pseudocons
tant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_relds (b 1) :required_relds (b 1) :incompatible_relds (b) :outer_relds (b) :le
ft_relds (b 1) :right_relds (b) :orclause <> :rinfo_serial 4 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies <> :left_em <> :right_em <> :outer_is_left fals
e :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hashseqoperator 0 :right_hashseqoperator 0} {RESTRICTINFO :clause {OPEXPR :opno 96 :opfuncid 65 :op
resulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args {(VAR :varno 1 :varattno 2 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varatt
nosyn 2 :location -1} {PARAM :paramkind 1 :paramid 0 :paramtype 23 :paramtypmod -1 :paramcollid 0 :location -1}) :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone fals
e :is_clone false :leakproof false :has_volatile 2 :security_level 0 :num_base_rels 1 :clause_relds (b 1) :required_relds (b 1) :incompatible_relds (b) :outer_relds (b) :left_relds (b 1) :right_rel
ids (b) :orclause <> :rinfo_serial 1 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies (o 1976) :left_em {EQUIVALENCEMEMBER :em_expr {(VAR :varno 1 :varattno 2
:vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1} :constvalue 4 [ -34 -8 -1 -1 -1 -1 -1 ]}) :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproo
f false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_relds (b 1) :required_relds (b 1) :incompatible_relds (b) :outer_relds (b) :left_relds (b 1) :right_relds (b) :orclause <> :rinfo_
serial 3 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_buc
ketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hashseqoperator 0 :right_hashseqoperator 0} {RESTRICTINFO :clause {OPEXPR :opno 96 :opfuncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args {(VAR :varno 1 :varattno 3 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1} :constvalue 4 [ -34 -8 -1 -1 -1 -1 -1 ]}) :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_relds
(b 1) :required_relds (b 1) :incompatible_relds (b) :outer_relds (b) :left_relds (b 1) :right_relds (b) :orclause <> :rinfo_serial 2 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :ou
ter_selec -1 :mergeopfamilies (o 1976) :left_em {EQUIVALENCEMEMBER :em_expr {(VAR :varno 1 :varattno 3 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1} :em_relds (b 1) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 1)}} :right_em {EQUIVALENCEMEMBER :em_expr {(PARAM :paramkind 1 :paramid 0 :paramtype 23 :paramtypmod -1 :paramcollid 0 :location -1} :em_relds (b 1) :em_is_const true :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 1)}} :outer_is_left fa
lse :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hashseqoperator 0 :right_hashseqoperator 0}} | parallel_safe = false | reason = PROPARALLEL_RESTRICTED
WARNING: max_parallel_hazard_walker: Param paramid=0 in_safe_list=0 => reason=outer_reference
WARNING: is_parallel_safe: OUTPUT qual = (({RESTRICTINFO :clause {OPEXPR :opno 1098 :opfuncid 1090 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args {(VAR :varno 1 :varattno 11 :vartype 1
082 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 11 :location -1} {CONST :consttype 1082 :consttypmod -1 :constcollid 0 :constlen 4 :constb
yval true :constisnull false :location -1 :constvalue 4 [ -34 -8 -1 -1 -1 -1 -1 ]}) :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproo
f false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_relds (b 1) :required_relds (b 1) :incompatible_relds (b) :outer_relds (b) :left_relds (b 1) :right_relds (b) :orclause <> :rinfo_
serial 3 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_buc
ketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hashseqoperator 0 :right_hashseqoperator 0} {RESTRICTINFO :clause {OPEXPR :opno 2345 :opfuncid 2338 :opresulttype 16 :opretset false :opcollid 0 :inputc
ollid 0 :args {(VAR :varno 1 :varattno 11 :vartype 1082 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 11 :location -1} {CONST :consttype 1114
:consttypmod -1 :constcollid 0 :constlen 8 :constbyval true :constisnull false :location -1 :constvalue 8 [ 0 32 -39 -79 49 -115 -1 -1 ]}) :location -1} :is_pushed_down true :can_join false :pseudocons
tant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_relds (b 1) :required_relds (b 1) :incompatible_relds (b) :outer_relds (b) :le
ft_relds (b 1) :right_relds (b) :orclause <> :rinfo_serial 4 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies <> :left_em <> :right_em <> :outer_is_left fals
e :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hashseqoperator 0 :right_hashseqoperator 0} {RESTRICTINFO :clause {OPEXPR :opno 96 :opfuncid 65 :op
resulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args {(VAR :varno 1 :varattno 2 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varatt
nosyn 2 :location -1} {PARAM :paramkind 1 :paramid 0 :paramtype 23 :paramtypmod -1 :paramcollid 0 :location -1}) :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone fals
e :is_clone false :leakproof false :has_volatile 2 :security_level 0 :num_base_rels 1 :clause_relds (b 1) :required_relds (b 1) :incompatible_relds (b) :outer_relds (b) :left_relds (b 1) :right_rel
ids (b) :orclause <> :rinfo_serial 1 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies (o 1976) :left_em {EQUIVALENCEMEMBER :em_expr {(VAR :varno 1 :varattno 2
:vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1} :constvalue 4 [ -34 -8 -1 -1 -1 -1 -1 ]}) :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproo
f false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_relds (b 1) :required_relds (b 1) :incompatible_relds (b) :outer_relds (b) :left_relds (b 1) :right_relds (b) :orclause <> :rinfo_
serial 3 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_buc
ketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hashseqoperator 0 :right_hashseqoperator 0} {RESTRICTINFO :clause {OPEXPR :opno 96 :opfuncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args {(VAR :varno 1 :varattno 3 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1} :constvalue 4 [ -34 -8 -1 -1 -1 -1 -1 ]}) :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_relds
(b 1) :required_relds (b 1) :incompatible_relds (b) :outer_relds (b) :left_relds (b) :orclause <> :rinfo_serial 2 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :ou
ter_selec -1 :mergeopfamilies (o 1976) :left_em {EQUIVALENCEMEMBER :em_expr {(VAR :varno 1 :varattno 3 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1} :em_relds (b 1) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 1)}} :right_em {EQUIVALENCEMEMBER :em_expr {(PARAM :paramkind 1 :paramid 0 :paramtype 23 :paramtypmod -1 :paramcollid 0 :location -1} :em_relds (b 1) :em_is_const true :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 1)}} :outer_is_left f
alse :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hashseqoperator 0 :right_hashseqoperator 0}} | parallel_safe = false | reason = PROPARALLEL_RESTRICTED
WARNING: is_parallel_safe: INPUT qual = (({OPEXPR :opno 1760 :opfuncid 1726 :opresulttype 1700 :opretset false :opcollid 0 :inputcollid 0 :args {(CONST :consttype 1700 :consttypmod -1 :constcollid 0 :c
onstlen 1 :constbyval false :constisnull false :location -1 :constvalue 8 [ 32 0 0 0 -1 -128 -128 19 ]} {AGGREF :aggfnoid 2114 :aggtype 1700 :aggcollid 0 :inputcollid 0 :aggtrantype 2281 :aggartypes (
o 1700) :aggdirectargs <> :args {(TARGETENTRY :expr {(VAR :varno 1 :varattno 5 :vartype 1700 :vartypmod 983846 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn
5 :location -1} :resno 1 :resname <> :ressortgroupref 0 :resorigtbl 0 :resorigcol 0 :resjunk false)} :aggorder <> :aggdistinct <> :aggfilter <> :aggstar false :aggvariadic false :aggkind n :aggpresorte
d false :agglevelsup 0 :aggssplit 0 :aggno 0 :aggtransno 0 :location -1}) :location -1}) :location -1} | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: OUTPUT qual = (({OPEXPR :opno 1760 :opfuncid 1726 :opresulttype 1700 :opretset false :opcollid 0 :inputcollid 0 :args {(CONST :consttype 1700 :consttypmod -1 :constcollid 0 :c
onstlen 1 :constbyval false :constisnull false :location -1 :constvalue 8 [ 32 0 0 0 -1 -128 -128 19 ]} {AGGREF :aggfnoid 2114 :aggtype 1700 :aggcollid 0 :inputcollid 0 :aggtrantype 2281 :aggartypes (
o 1700) :aggdirectargs <> :args {(TARGETENTRY :expr {(VAR :varno 1 :varattno 5 :vartype 1700 :vartypmod 983846 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn
n 5 :location -1} :resno 1 :resname <> :ressortgroupref 0 :resorigtbl 0 :resorigcol 0 :resjunk false)} :aggorder <> :aggdistinct <> :aggfilter <> :aggstar false :aggvariadic false :aggkind n :aggpresorte
d false :agglevelsup 0 :aggssplit 0 :aggno 0 :aggtransno 0 :location -1}) :location -1}) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = (({VAR :varno 1 :varattno 5 :vartype 1700 :vartypmod 983846 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 5 :locati
on -1}) :location -1} | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: Log 2.2 After pull up At starting of Path Generation:
DETAIL:
{
  QUERY
  {
    commandType 1
    querySource 0
    canSetTag true
    utilityStmt <>
    resultRelation 0
    hasAggs false
    hasWindowFuncs false
    hasTargetSRFs false
    hasSubLinks true
    hasDistinctOn false
    hasRecursive false
    hasModifyingCTE false
    hasForUpdate false
    hasRowSecurity false
    hasGroupRTE false
    isReturn false
    cteList <>
  }
  {
    table 0
    {
      {RANGETBLENTY
      :
      :alias <>
      :eref
      {
        {ALIAS
        :
        :aliasname supplier
        :colnames ("s_supplier" "s_name" "s_address" "s_nationkey" "s_phone" "
s_acctbal" "s_comment")
        }
      }
      :rtekind 0
      :relid 17399
      :inh false
      :relkind r
      :rellockmode 1
      :perminfoindex 1
      :tablesample <>
      :lateral false
      :inFromC1 true
      :securityQuals <>
      }
    }
    {RANGETBLENTY
    :
    :alias <>
    :eref
    {
      {ALIAS
      :
      :aliasname ANY_subquery
      :colnames <>
      }
    }
    :rtekind 1
    :subquery <>
    :security_barrier false
    :relid 0
    :inh false
    :relkind <>
    :rellockmode 0
    :perminfoindex 0
    :lateral false
    :inFromC1 false
    :securityQuals <>
  }
}
```

```

}
{RANGETBLENTY
:alias <>
:eref
{ALIAS
:aliasname partsupp_secure
:colnames ("ps_partkey" "ps_supkey" "ps_availqty" "ps_supplycost" "p
s_comment")
}
}
:rtekind 1
:subquery
{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks true
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable (
{RANGETBLENTY
:alias
{ALIAS
:aliasname ps
:colnames <>
}
}
:eref
{ALIAS
:aliasname ps
:colnames ("ps_partkey" "ps_supkey" "ps_availqty" "ps_supplyco
st" "ps_comment")
}
}
:rtekind 0
:relid 17409
:inh true
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCI true
:securityQuals <>
}
)
:rteperminfos (
{RTEPERMISSIONINFO
:relid 17409
:inh true
:requiredPerms 2
:checkAsUser 10
:selectedCols (b 8 9 10 11 12)
:insertedCols (b)
:updatedCols (b)
}
)
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
)
}
:quals
{BOOLEXP
:boolop and
:args (
{OPEXP
:opno 521
:opfuncid 147
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 3
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 3
:location -1
}
)
{CONST
:consttype 23
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location -1
:constvalue 4 [ 100 0 0 0 0 0 0 ]
}
)
:location -1
}
)
{SUBLINK
:subLinkType 0
:subLinkId 0
:testexpr <>
:operName <>
:subselect
{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable (
{RANGETBLENTY
:alias
{ALIAS
:aliasname s
:colnames <>
}
}
:eref
{ALIAS
:aliasname s
:colnames ("s_supkey" "s_name" "s_address" "s_nati
onkey" "s_phone" "s_acctbal" "s_comment")
}
}
:rtekind 0

```

```

:relid 17399
:inh true
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
)
:rteperminfos (
{RTEPERMISSIONINFO
:relid 17399
:inh true
:requiredPerms 2
:checkAsUser 10
:selectedCols (b 8 13)
:insertedCols (b)
:updatedCols (b)
}
)
):jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
)
):quals
{BOOLEXP
:boolop and
:args (
{OEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location -1
}
{VAR
:varno 1
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 1
:varreturningtype 0
:varnosyn 1
:varattnosyn 2
:location -1
}
)
:location -1
}
{OEXPR
:opno 1756
:opfuncid 1720
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 6
:vartype 1700
:vartypmod 983046
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location -1
}
{FUNCEXP
:funcid 1740
:funcresulttype 1700
:funcresultset false
:funcvariadic false
:funcformat 2
:funccollid 0
:inputcollid 0
:args (
{CONST
:consttype 23
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location -1
:constvalue 4 [ 0 0 0 0 0 0 0 ]
}
)
:location -1
}
)
:location -1
}
)
:location -1
}
):mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetlist (
{TARGETENTRY
:expr
{CONST
:consttype 23
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location -1
:constvalue 4 [ 1 0 0 0 0 0 0 ]
}
}
:resno 1
:resname ?column?
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
):override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>

```

```

        :returningList <>
        :groupClause <>
        :groupDistinct false
        :groupingSets <>
        :havingQual <>
        :windowClause <>
        :distinctClause <>
        :sortClause <>
        :limitOffset <>
        :limitCount <>
        :limitOption 0
        :rowMarks <>
        :setOperations <>
        :constraintDeps <>
        :withCheckOptions <>
        :stmt_location -1
        :stmt_len -1
    }
    :location -1
}
}
:location -1
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 1
:varatype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location -1
}
:resno 1
:resname ps_partkey
:ressortgroupref 0
:resorigtbl 17409
:resorigcol 1
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 2
:varatype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 2
:location -1
}
}
:resno 2
:resname ps_suppkey
:ressortgroupref 0
:resorigtbl 17409
:resorigcol 2
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 3
:varatype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 3
:location -1
}
}
:resno 3
:resname ps_availqty
:ressortgroupref 0
:resorigtbl 17409
:resorigcol 3
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 4
:varatype 1700
:vartypmod 983046
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 4
:location -1
}
}
:resno 4
:resname ps_supplycost
:ressortgroupref 0
:resorigtbl 17409
:resorigcol 4
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 5
:varatype 1043
:vartypmod 203
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 5
:location -1
}
}
:resno 5
:resname ps_comment
:ressortgroupref 0
:resorigtbl 17409
:resorigcol 5
:resjunk false
}
}
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>

```

```

:windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}
:security_barrier true
:relid 19883
:inh false
:relkind v
:rellockmode 1
:perminfoindex 3
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBLENTRY
:alias
{ALIAS
:aliasname ANY_subquery
:colnames <>
}
}
:eref
{ALIAS
:aliasname ANY_subquery
:colnames ("p_partkey")
}
}
:rtekind 1
:subquery <>
:security_barrier false
:relid 0
:inh false
:relkind <>
:rellockmode 0
:perminfoindex 0
:lateral false
:inFromCl false
:securityQuals <>
}
{RANGETBLENTRY
:alias <>
:eref
{ALIAS
:aliasname part
:colnames ("p_partkey" "p_name" "p_mfg" "p_brand" "p_type" "p_size"
"p_container" "p_retailprice" "p_comment")
}
}
:rtekind 0
:relid 17387
:inh false
:relkind r
:rellockmode 1
:perminfoindex 4
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
}
)
:rteperminfos (
{RTEPERMISSIONINFO
:relid 17399
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9 10 11)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 17376
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 19883
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9 10)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 17387
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9)
:insertedCols (b)
:updatedCols (b)
}
)
}
:jointree
{FROMEXPR
:fromlist (
{JOINEXPR
:jointype 4
:isNatural false
:larg
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
{RANGETBLREF
:rtindex 2
}
)
}
:quals (
{OEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 4
:varatype 23
:varitypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 4
:location 1240
}
{VAR
:varno 2
:varattno 1
:varatype 23
:varitypmod -1
:varcollid 0

```

```

:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 1
:location 1254
}
)
:location 1252
}
{OPEXPR
:opno 1054
:opfuncid 1048
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 100
:args (
{VAR
:varno 2
:varattno 2
:vartype 1042
:vartypmod 29
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 2
:location 1278
}
{CONST
:consttype 1042
:consttypmod -1
:constcollid 100
:constlen -1
:constbyval false
:constisnull false
:location 1287
:constvalue 10 [ 40 0 0 0 70 82 65 78 67 69 ]
}
)
:location 1285
}
)
}
:rarg
{JOINEXPR
:jointype 4
:isNatural false
:larg
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 4
}
)
:quals (
{OPEXPR
:opno 1756
:opfuncid 1720
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{FUNCEXPR
:funcid 1740
:funcresulttype 1700
:funcsetset false
:funcvariadic false
:funcformat 2
:funccollid 0
:inputcollid 0
:args (
{VAR
:varno 4
:varattno 3
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 3
:location 650
}
)
:location -1
}
{SUBPLAN
:sublinkType 4
:testexpr <>
:paramIds <>
:plan_id 1
:plan_name SubPlan\ 1
:firstColType 1700
:firstColTypmod -1
:firstColCollation 0
:useHashTable false
:unknownEqFalse false
:parallel_safe false
:setParam <>
:parParam (i 0 1)
:args (
{VAR
:varno 4
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 1
:location 943
}
{VAR
:varno 4
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 2
:location 1010
}
)
:startup_cost 0
:per_call_cost 8.469999999999999
}
)
:location 662
}
)
}
:rarg
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 6
}
)
)

```

```

:quals (
{OPEXPR
:opno 1209
:opfuncid 850
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 100
:args (
{RELABELTYPE
:arg
{VAR
:varno 6
:varattno 2
:vartype 1043
:vartypmod 59
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 6
:varattnosyn 2
:location 574
}
:resulttype 25
:resulttypmod -1
:resultcollid 100
:relabelformat 2
:location -1
}
{CONST
:consttype 25
:consttypmod -1
:constcollid 100
:constlen -1
:constbyval false
:constisnull false
:location 586
:constvalue 11 [ 44 0 0 0 37 105 118 111 114 121 37 ]
}
}
:location 581
}
)
}
:usingClause <>
:join_using_alias <>
:quals (
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 4
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 1
:location 309
}
{VAR
:varno 6
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 6
:varattnosyn 1
:location 404
}
}
:location 320
}
)
:alias <>
:rtindex 0
}
:usingClause <>
:join_using_alias <>
:quals (
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location 129
}
{VAR
:varno 4
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 2
:location 191
}
}
:location 139
}
)
:alias <>
:rtindex 0
}
)
:quals <>
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 2
:vartype 1042
:vartypmod 29
:varcollid 100

```

```
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 2
:location 51
}
:resno 1
:resname s_name
:ressortgroupref 1
:resorigtbl 17399
:resorigcol 2
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 3
:vartype 1043
:vartypmod 44
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 3
:location 67
}
:resno 2
:resname s_address
:ressortgroupref 0
:resorigtbl 17399
:resorigcol 3
:resjunk false
}
}
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupdistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause (
{SORTGROUPCLAUSE
:tsSortGroupRef 1
:eqop 1054
:sortop 1058
:reverse_sort false
:nulls_first false
:hashable true
}
)
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDecls <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 0
}

WARNING: is_parallel_safe: INPUT qual = <>
WARNING: is_parallel_safe: OUTPUT qual = <> | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ({(VAR :varno 1 :varattno 2 :vartype 1042 :vartypmod 29 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 2 :location -1) (VAR :varno 1 :varattno 3 :vartype 1043 :vartypmod 44 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1) (VAR :varno 1 :varattno 4 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 4 :location -1) (VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1)})
WARNING: is_parallel_safe: OUTPUT qual = ({(VAR :varno 1 :varattno 2 :vartype 1042 :vartypmod 29 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 2 :location -1) (VAR :varno 1 :varattno 3 :vartype 1043 :vartypmod 44 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1) (VAR :varno 1 :varattno 4 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 4 :location -1) (VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1)})
WARNING: is_parallel_safe: INPUT qual = ({(RESTRICTINFO :clause {OEXPR :opno 1054 :opfuncid 1048 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 100 :args ({(VAR :varno 2 :varattno 2 :vartype 1042 :vartypmod 29 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 2 :location -1) :em_relds (b 2) :em_is_const false :em_is_child false :em_datatype 1042 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2 4 6)}) :right_em {EQUIVALENCEMEMBER :em_expr {CONST :consttype 1042 :consttypmod -1 :constcollid 100 :constlen -1 :constbyval false :constisnull false :location -1 :constvalue 10 [ 4 0 0 0 70 82 65 78 67 69 ]} :em_relds (b) :em_is_const true :em_is_child false :em_datatype 1042 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2 4 6)}) :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hashoperator 0 :right_hashoperator 0}) :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hashoperator 0 :right_hashoperator 0})
WARNING: is_parallel_safe: OUTPUT qual = ({(RESTRICTINFO :clause {OEXPR :opno 1054 :opfuncid 1048 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 100 :args ({(VAR :varno 2 :varattno 2 :vartype 1042 :vartypmod 29 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 2 :location -1) :em_relds (b 2) :em_is_const false :em_is_child false :em_datatype 1042 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2 4 6)}) :right_em {EQUIVALENCEMEMBER :em_expr {CONST :consttype 1042 :consttypmod -1 :constcollid 100 :constlen -1 :constbyval false :constisnull false :location -1 :constvalue 10 [ 4 0 0 0 70 82 65 78 67 69 ]} :em_relds (b) :em_is_const true :em_is_child false :em_datatype 1042 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2 4 6)}) :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hashoperator 0 :right_hashoperator 0}) :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hashoperator 0 :right_hashoperator 0})
WARNING: is_parallel_safe: INPUT qual = ({(VAR :varno 2 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 1 :location -1)})
WARNING: is_parallel_safe: OUTPUT qual = ({(VAR :varno 2 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 1 :location -1)})
WARNING: is_parallel_safe: INPUT qual = ({(RESTRICTINFO :clause {OEXPR :opno 1756 :opfuncid 1720 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({(FUNCEXP :funcid 1740 :funcresulttype 0 :varnosyn 4 :varattnosyn 3 :location -1) :location -1) (SUBPLAN :subLinkType 4 :testexpr <> :paramIds <> :plan_id 1 :plan_name SubPlan\ 1 :firstColType 1700 :firstColTypepmod -1 :firstColCollation 0 :useHashTable false :unknownEq false :parallel_safe false :setParam <> :parParam (i 0 1) :args ({(VAR :varno 4 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 4 :varattnosyn 1 :location -1) :location -1) (VAR :varno 4 :varattno 2 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 4 :varattnosyn 2 :location -1) :location -1) (VAR :varno 4 :varattno 3 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 4 :varattnosyn 3 :location -1) :location -1) (VAR :varno 4 :varattno 4 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 4 :varattnosyn 4 :location -1) :location -1) :start_cost 0 :per_call_cost 8.469999999999999 :location -1) :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_relds (b 2) :required_relds (b 2) :incompatible_relds (b 2) :outer_relds (b 2) :left_relds (b 2) :right_relds (b 2) :orclause <> :rinfo_serial 3 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hashoperator 0 :right_hashoperator 0}) :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hashoperator 0 :right_hashoperator 0})
WARNING: max_parallel_hazard_walker: FuncExpr funcid=1740 proparallel=s (s=safe r=restricted u=unsafe)
WARNING: max_parallel_hazard_walker: SubPlan plan_id=1 parallel_safe=0 + reason=correlated_subplan
WARNING: is_parallel_safe: OUTPUT qual = ({(RESTRICTINFO :clause {OEXPR :opno 1756 :opfuncid 1720 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({(FUNCEXP :funcid 1740 :funcresulttype 0 :varnosyn 4 :varattnosyn 3 :location -1) :location -1) (SUBPLAN :subLinkType 4 :testexpr <> :paramIds <> :plan_id 1 :plan_name SubPlan\ 1 :firstColType 1700 :firstColTypepmod -1 :firstColCollation 0 :useHashTable false :unknownEq false :parallel_safe false :setParam <> :parParam (i 0 1) :args ({(VAR :varno 4 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 4 :varattnosyn 1 :location -1) :location -1) (VAR :varno 4 :varattno 2 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 4 :varattnosyn 2 :location -1) :location -1) (VAR :varno 4 :varattno 3 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 4 :varattnosyn 3 :location -1) :location -1) (VAR :varno 4 :varattno 4 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 4 :varattnosyn 4 :location -1) :location -1) :start_cost 0 :per_call_cost 8.469999999999999 :location -1) :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_relds (b 2) :required_relds (b 2) :incompatible_relds (b 2) :outer_relds (b 2) :left_relds (b 2) :right_relds (b 2) :orclause <> :rinfo_serial 3 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hashoperator 0 :right_hashoperator 0}) :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hashoperator 0 :right_hashoperator 0})
WARNING: Log 2: After pull up At starting of Path Generation:
DETAIL: (QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks true
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:ctelist <>
:table (
{RANGEtblENTRY
:alias
{ALIAS
:aliasname ps
:colnames <>
}
}
:eref
```

```
ps_availqty > (
SELECT 0.5 * sum(l_quantity)
FROM lineitem
WHERE l_partkey = ps_partkey
AND l_suppkey = ps_suppkey
AND l_shipdate >= DATE '1995-01-01'
AND l_shipdate < DATE '1995-01-01' + INTERVAL '1 year'
)
```

```

{ALIAS
:aliasname ps
:colnames ("ps_partkey" "ps_suppkey" "ps_availqty" "ps_supplycost" "p
s_comment")
}
:rtekind 0
:relid 17409
:inh false
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBLENTRY
:alias
{ALIAS
:aliasname s
:colnames <>
}
}
:eref
{ALIAS
:aliasname s
:colnames ("s_suppkey" "s_name" "s_address" "s_nationkey" "s_phone" "
s_acctbal" "s_comment")
}
:rtekind 0
:relid 17399
:inh false
:relkind r
:rellockmode 1
:perminfoindex 2
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
}
:rteperminfos (
{RTEPERMISSIONINFO
:relid 17409
:inh true
:requiredPerms 2
:checkAsUser 10
:selectedCols (b 8 9 10 11 12)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 17399
:inh true
:requiredPerms 2
:checkAsUser 10
:selectedCols (b 8 13)
:insertedCols (b)
:updatedCols (b)
}
)
:jointree
{FROMEXPR
:fromlist (
{JOINEXPR
:jointype 4
:isNatural false
:larg
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
)
:quals (
{OPEXPR
:opno 521
:opfuncid 147
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 3
:vartype 23
:varitypmode -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 3
:location -1
}
)
{CONST
:consttype 23
:constitypmode -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location -1
:constvalue 4 [ 100 0 0 0 0 0 0 0 ]
}
)
:location -1
}
}
}
:rarg
{RANGETBLREF
:rtindex 2
}
:usingClause <>
:join_using_alias <>
:quals (
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 2
:varattno 1
:vartype 23
:varitypmode -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 1
:location -1
}
)
{VAR
:varno 1
:varattno 2
:vartype 23
:varitypmode -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
}
}
}

```

```

        :varattnosyn 2
        :location -1
    }
}
:location -1
}
{OPEXPR
:opno 1756
:opfuncid 1720
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 2
:varattno 6
:vartype 1700
:vartypmod 983046
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 6
:location -1
}
{CONST
:consttype 1700
:consttypmod -1
:constcollid 0
:constlen -1
:constbyval false
:constisnull false
:location -1
:constvalue 6 [ 24 0 0 0 0 -128 ]
}
)
:location -1
}
)
:alias <>
:rtindex 0
}
)
:quals <>
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location -1
}
:resno 1
:resname ps_partkey
:ressortgroupref 0
:resorigtbl 17409
:resorigcol 1
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 2
:location -1
}
:resno 2
:resname ps_supplekey
:ressortgroupref 0
:resorigtbl 17409
:resorigcol 2
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 3
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 3
:location -1
}
:resno 3
:resname ps_availqty
:ressortgroupref 0
:resorigtbl 17409
:resorigcol 3
:resjunk false
}
{TARGETENTRY
:expr
{CONST
:consttype 1700
:consttypmod 983046
:constcollid 0
:constlen -1
:constbyval false
:constisnull true
:location -1
:constvalue <>
}
:resno 4
:resname ps_supplycost
:ressortgroupref 0
:resorigtbl 17409
:resorigcol 4
:resjunk false
}
{TARGETENTRY
:expr
{CONST
:consttype 1043
:consttypmod 203
:constcollid 100
:constlen -1
:constbyval false
:constisnull true
:location -1
:constvalue <>
}
}

```

```

:resno 5
:resname ps.comment
:ressortgroupref 0
:resorigtbl 17409
:resorigcol 5
:resjunk false
}

:override 0
:conflict <>
:returingOldAlias <>
:returingNewAlias <>
:returingList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
:windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}

WARNING: is_parallel_safe: INPUT qual = ((RESTRICTINFO :clause (OPEXPR :opno 521 :opfuncid 147 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ((VAR :varno 1 :varattno 3 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1) (CONST :consttype 23 :constypmod -1 :constcollid 0 :constlen 4 :constbval true :constisnull false :location -1 :constvalue 4 [ 100 0 0 0 0 0 0 0 ])) :location -1) :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_relds (b 1) :required_relds (b 1) :incompatible_relds (b) :outer_relds (b) :left_relds (b 1) :right_relds (b) :orclause <> :rinfo_serial 1 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_select -1 :outer_select -1 :mergeofamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mvcfreq -1 :right_mvcfreq -1 :left_hashoperator 0 :right_hashoperator 0))
WARNING: is_parallel_safe: OUTPUT qual = ((RESTRICTINFO :clause (OPEXPR :opno 521 :opfuncid 147 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ((VAR :varno 1 :varattno 3 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1) (CONST :consttype 23 :constypmod -1 :constcollid 0 :constlen 4 :constbval true :constisnull false :location -1 :constvalue 4 [ 100 0 0 0 0 0 0 0 ])) :location -1) :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_relds (b 1) :required_relds (b 1) :incompatible_relds (b) :outer_relds (b) :left_relds (b 1) :right_relds (b) :orclause <> :rinfo_serial 1 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_select -1 :outer_select -1 :mergeofamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mvcfreq -1 :right_mvcfreq -1 :left_hashoperator 0 :right_hashoperator 0))
WARNING: is_parallel_safe: INPUT qual = ((VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1) (VAR :varno 1 :varattno 2 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 2 :location -1) (VAR :varno 1 :varattno 3 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1))
WARNING: is_parallel_safe: OUTPUT qual = ((VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1) (VAR :varno 1 :varattno 2 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 2 :location -1) (VAR :varno 1 :varattno 3 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1))
WARNING: is_parallel_safe: INPUT qual = ((RESTRICTINFO :clause (OPEXPR :opno 1756 :opfuncid 1720 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ((VAR :varno 2 :varattno 6 :vartype 17 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 6 :location -1) (CONST :consttype 1700 :constypmod -1 :constcollid 0 :constlen -1 :constbval false :constisnull false :location -1 :constvalue 6 [ 24 0 0 0 0 -128 ])) :location -1) :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 2 :clause_relds (b 2) :required_relds (b 2) :incompatible_relds (b) :outer_relds (b) :left_relds (b 2) :right_relds (b) :orclause <> :rinfo_serial 1 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_select -1 :outer_select -1 :mergeofamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mvcfreq -1 :right_mvcfreq -1 :left_hashoperator 0 :right_hashoperator 0))
WARNING: is_parallel_safe: OUTPUT qual = ((RESTRICTINFO :clause (OPEXPR :opno 1756 :opfuncid 1720 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ((VAR :varno 2 :varattno 6 :vartype 17 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 6 :location -1) (CONST :consttype 1700 :constypmod -1 :constcollid 0 :constlen -1 :constbval false :constisnull false :location -1 :constvalue 6 [ 24 0 0 0 0 -128 ])) :location -1) :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 2 :clause_relds (b 2) :required_relds (b 2) :incompatible_relds (b) :outer_relds (b) :left_relds (b 2) :right_relds (b) :orclause <> :rinfo_serial 1 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_select -1 :outer_select -1 :mergeofamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mvcfreq -1 :right_mvcfreq -1 :left_hashoperator 0 :right_hashoperator 0))
WARNING: is_parallel_safe: INPUT qual = ((VAR :varno 2 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 1 :location -1) (VAR :varno 2 :varattno 2 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 2 :location -1) (VAR :varno 2 :varattno 3 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 3 :location -1))
WARNING: is_parallel_safe: OUTPUT qual = ((VAR :varno 2 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 1 :location -1) (VAR :varno 2 :varattno 2 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 2 :location -1) (VAR :varno 2 :varattno 3 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 3 :location -1))
WARNING: is_parallel_safe: INPUT qual = ((RESTRICTINFO :clause (OPEXPR :opno 96 :opfuncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ((VAR :varno 1 :varattno 2 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 2 :location -1) (CONST :consttype 1700 :constypmod -1 :constcollid 0 :constlen -1 :constbval false :constisnull false :location -1 :constvalue <>)) :location -1) :is_pushed_down true :can_join true :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 2 :security_level 0 :num_base_rels 2 :clause_relds (b 1 2) :required_relds (b 1 2) :incompatible_relds (b) :outer_relds (b 1) :right_relds (b 2) :orclause <> :rinfo_serial 4 :eval_cost.startup 0 :eval_cost.per_tuple 0.0025 :norm_select -1 :outer_select 0.9005 :mergeofamilies (o 1976) :left_em (EQUIVALENCEMEMBER :em_expr (VAR :varno 1 :varattno 2 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 2 :location -1) :em_relds (b 1) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain (JOINDOMAIN :jd_relds (b 1 2))) :right_em (EQUIVALENCEMEMBER :em_expr (VAR :varno 2 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 1 :location -1) :em_relds (b 2) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain (JOINDOMAIN :jd_relds (b 1 2))) :outer_is_left false :hashjoinoperator 96 :left_bucketsize -1 :right_bucketsize -1 :left_mvcfreq -1 :right_mvcfreq -1 :left_hashoperator 96 :right_hashoperator 96))
WARNING: is_parallel_safe: OUTPUT qual = ((RESTRICTINFO :clause (OPEXPR :opno 96 :opfuncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ((VAR :varno 1 :varattno 2 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 2 :location -1) (CONST :consttype 1700 :constypmod -1 :constcollid 0 :constlen -1 :constbval false :constisnull false :location -1 :constvalue <>)) :location -1) :is_pushed_down true :can_join true :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 2 :security_level 0 :num_base_rels 2 :clause_relds (b 1 2) :required_relds (b 1 2) :incompatible_relds (b) :outer_relds (b 1) :right_relds (b 2) :orclause <> :rinfo_serial 4 :eval_cost.startup 0 :eval_cost.per_tuple 0.0025 :norm_select -1 :outer_select 0.9005 :mergeofamilies (o 1976) :left_em (EQUIVALENCEMEMBER :em_expr (VAR :varno 1 :varattno 2 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 2 :location -1) :em_relds (b 1) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain (JOINDOMAIN :jd_relds (b 1 2))) :right_em (EQUIVALENCEMEMBER :em_expr (VAR :varno 2 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 1 :location -1) :em_relds (b 2) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain (JOINDOMAIN :jd_relds (b 1 2))) :outer_is_left false :hashjoinoperator 96 :left_bucketsize -1 :right_bucketsize -1 :left_mvcfreq -1 :right_mvcfreq -1 :left_hashoperator 96 :right_hashoperator 96))
```

artypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 4 :location -1} {VAR :varno 2 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 1 :location -1} :location -1} :is_pushed_down true :can_join true :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 2 :security_level 0 :num_base_rels 2 :clause_relds (b 1 2) :required_relds (b 1 2) :incompatible_relds (b) :outer_relds (b 1) :right_relds (b 2) :orclause <> :rinfo_ serial 7 :eval_cost :startup 0 :eval_cost_per_tuple 0.0025 :norm_selec 0.04 :outer_selec -1 :mergeofamilies (0 1976) :left_em (EQUIVALENCEMEMBER :em_expr (VAR :varno 1 :varattno 4 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 4 :location -1) :em_relds (b 1) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOIND OMAIN :jd_relds (b 1 2 4 6)}) :right_em (EQUIVALENCEMEMBER :em_expr (VAR :varno 2 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 2 :varattnosyn 1 :location -1) :em_relds (b 2) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOIND OMAIN :jd_relds (b 1 2 4 6)}) :outer_is_left false :hashjoinoperator 96 :left_bucket size -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 96 :right_hasheqoperator 96)) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = {(VAR :varno 1 :varattno 2 :vartype 1042 :vartypmod 29 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 2 :location -1} (VAR :varno 1 :varattno 3 :vartype 1043 :vartypmod 44 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1)} | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: OUTPUT qual = {(VAR :varno 1 :varattno 2 :vartype 1042 :vartypmod 29 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1) :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1} (VAR :varno 1 :varattno 1 :vartype 1043 :vartypmod 44 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 2 :location -1)} | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = {(VAR :varno 1 :varattno 2 :vartype 1042 :vartypmod 29 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 2 :location -1) (VAR :varno 1 :varattno 3 :vartype 1043 :vartypmod 44 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1)} | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: OUTPUT qual = {(VAR :varno 1 :varattno 2 :vartype 1042 :vartypmod 29 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 2 :location -1) (VAR :varno 1 :varattno 3 :vartype 1043 :vartypmod 44 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1)} | parallel_safe = true | reason = PROPARALLEL_SAFE
n = PROPARALLEL_SAFE

QUERY PLA

N

Sort (cost=1222175.01..1222175.11 rows=40 width=52) (actual time=2342.207..2342.212 rows=65.00 loops=1)
Output: supplier.s_name, supplier.s_address
Sort Key: supplier.s_name
Sort Method: quicksort Memory: 29kB
Buffers: shared hit=545369 read=48654 written=22
-> Nested Loop Semi Join (cost=1132.37..1222173.95 rows=40 width=52) (actual time=1660.984..2341.863 rows=65.00 loops=1)
Output: supplier.s_name, supplier.s_address
Join Filter: (supplier.s_supkey = partsupp_secure.ps_supkey)
Rows Removed by Join Filter: 161820
Buffers: shared hit=545366 read=48654 written=22
-> Nested Loop (cost=4.90..97.57 rows=80 width=56) (actual time=0.140..0.354 rows=78.00 loops=1)
Output: supplier.s_name, supplier.s_address, supplier.s_supkey
Buffers: shared hit=5 read=35
-> Seq Scan on public.nation (cost=0.00..1.31 rows=1 width=4) (actual time=0.064..0.085 rows=1.00 loops=1)
Output: nation.n_nationkey, nation.n_name, nation.n_regionkey, nation.n_comment
Filter: (nation.n_name = 'FRANCE'::bpchar)
Rows Removed by Filter: 24
Buffers: shared hit=1
-> Bitmap Heap Scan on public.supplier (cost=4.90..95.45 rows=80 width=60) (actual time=0.072..0.253 rows=78.00 loops=1)
Output: supplier.s_supkey, supplier.s_name, supplier.s_address, supplier.s_nationkey, supplier.s_phone, supplier.s_acctbal, supplier.s_comment
Recheck Cond: (supplier.s_nationkey = nation.n_nationkey)
Heap Blocks: exact=27
Buffers: shared hit=4 read=35
-> Bitmap Index Scan on idx_supplier_nationkey (cost=0.00..4.88 rows=80 width=0) (actual time=0.040..0.040 rows=78.00 loops=1)
Index Cond: (supplier.s_nationkey = nation.n_nationkey)
Index Searches: 1
Buffers: shared hit=2
-> Materialize (cost=1127.47..1218829.26 rows=3841 width=4) (actual time=0.268..29.908 rows=2075.45 loops=78)
Output: partsupp_secure.ps_supkey
Storage: Memory Maximum Storage: 264kB
Buffers: shared hit=545361 read=48619 written=22
-> Hash Semi Join (cost=1127.47..1218810.06 rows=3841 width=4) (actual time=20.899..2326.194 rows=5104.00 loops=1)
Output: partsupp_secure.ps_supkey
Hash Cond: (partsupp_secure.ps_partkey = part.p_partkey)
Buffers: shared hit=545361 read=48619 written=22
-> Subquery Scan on partsupp_secure (cost=0.93..1217651.00 rows=47541 width=8) (actual time=4.422..2305.548 rows=97475.00 loops=1)
Output: partsupp_secure.ps_partkey, partsupp_secure.ps_availqty, partsupp_secure.ps_supplycost, partsupp_secure.ps_comment
Filter: ((partsupp_secure.ps_availqty)::numeric > (SubPlan 1))
Rows Removed by Filter: 45196
Buffers: shared hit=544599 read=48619 written=22
-> Hash Semi Join (cost=135.93..7494.84 rows=142623 width=446) (actual time=2.491..55.910 rows=142671.00 loops=1)
Output: ps.ps_partkey, ps.ps_supkey, ps.ps_availqty, NULL::numeric(15,2), NULL::character varying(199)
Hash Cond: (ps.ps_supkey = s.s_supkey)
Buffers: shared hit=95 read=613 written=1
-> Index Only Scan using idx_partsupp_partkey_supkey_availqty on public.partsupp ps (cost=0.42..5356.89 rows=158382 width=12) (actual time=0.200..21.666 rows=158344.00 loops=1)
Output: ps.ps_partkey, ps.ps_supkey, ps.ps_availqty
Index Cond: (ps.ps_availqty > 100)
Heap Fetches: 0
Index Searches: 1
Buffers: shared hit=7 read=613 written=1
-> Hash (cost=113.00..113.00 rows=1801 width=4) (actual time=2.248..2.248 rows=1802.00 loops=1)
Output: s.s_supkey
Buckets: 2048 Batches: 1 Memory Usage: 80kB
Buffers: shared hit=80
-> Seq Scan on public.supplier s (cost=0.00..113.00 rows=1801 width=4) (actual time=0.020..1.707 rows=1802.00 loops=1)
Output: s.s_supkey
Filter: (s.s_acctbal > '0'::numeric)
Rows Removed by Filter: 198
Buffers: shared hit=88
SubPlan 1
-> Aggregate (cost=8.46..8.47 rows=1 width=32) (actual time=0.015..0.015 rows=1.00 loops=142671)
Output: (0.5 * sum(lineitem.l_quantity))
Buffers: shared hit=544504 read=48006 written=21
-> Index Scan using idx_lineitem_partkey_supkey_shipdate on public.lineitem (cost=0.43..8.45 rows=1 width=5) (actual time=0.010..0.015 rows=1.15 loops=142671)
Output: lineitem.l_orderkey, lineitem.l_partkey, lineitem.l_supplykey, lineitem.l_linenum, lineitem.l_quantity, lineitem.l_extendedprice, lineitem.l_discount, lineitem.l_tax, lineitem.l_returnflag, lineitem.l_linestatus, lineitem.l_shipdate, lineitem.l_commitdate, lineitem.l_receiptdate, lineitem.l_shipinstruct, lineitem.l_shipmode, lineitem.l_comment
Index Cond: ((lineitem.l_partkey = partsupp_secure.ps_partkey) AND (lineitem.l_supplykey = partsupp_secure.ps_supkey) AND (lineitem.l_shipdate >= '1995-01-01'::date) AND (lineitem.l_shipdate < '1996-01-01 00:00:00'::timestamp without time zone))
Index Searches: 142671
Buffers: shared hit=544504 read=48006 written=21
-> Hash (cost=951.14..951.14 rows=3232 width=4) (actual time=7.406..7.406 rows=2082.00 loops=1)
Output: part.p_partkey
Buckets: 4096 Batches: 1 Memory Usage: 106kB
Buffers: shared hit=762
-> Bitmap Heap Scan on public.part (cost=46.74..951.14 rows=3232 width=4) (actual time=1.889..6.865 rows=2082.00 loops=1)
Output: part.p_partkey
Recheck Cond: ((part.p_name)::text ~ 'Ivory%')::text)
Heap Blocks: exact=752
Buffers: shared hit=762
-> Bitmap Index Scan on idx_part_name_trgm (cost=0.00..45.93 rows=3232 width=0) (actual time=1.600..1.600 rows=2082.00 loops=1)
Index Cond: ((part.p_name)::text ~ 'Ivory%')::text)
Index Searches: 1
Buffers: shared hit=10
Planning:
Buffers: shared hit=393
Planning Time: 20.070 ms
Execution Time: 2343.536 ms (85 rows)
tpch-200mb=>